

pandas Problems

Level 1: Pandas Basics

1. Create a DataFrame

Ye dictionary di hui hai:

```
data = {  
    "Name": ["Amit", "Priya", "Rahul"],  
    "Age": [25, 28, 30],  
    "Salary": [50000, 60000, 70000]  
}
```

DataFrame banao.

2. Display Information

Upar wale DataFrame ke liye:

First 2 rows

Last row

Column names

Shape

nikalo.

3. Select Columns

Print:

Name

Salary

columns.

4. Select Single Row

Rahul wali row nikalo using:

loc

iloc

5. Add New Column

Ek new column banao:

Bonus = Salary * 0.10

6. Rename Column

Salary

ko

Monthly_Salary

rename karo.

7. Delete Column

Bonus column hatao.

Level 2: Filtering

8. Salary Greater Than 60000

Sirf un employees ko dikhao jinki salary 60000 se zyada hai.

9. Multiple Conditions

Age > 25 AND Salary > 55000

wale employees dikhao.

10. String Filtering

Sirf un logon ko dikhao jinke naam me letter:

"a"

aata ho.

11. Top Earners

Highest salary wale employee ko nikalo.

Level 3: Sorting

12. Sort by Salary

Salary ascending order me.

13. Sort by Age

Age descending order me.

14. Multiple Column Sorting

Age ascending aur Salary descending.

Level 4: Missing Values

Dataset:

```
data = {  
    "Name": ["Amit", "Priya", "Rahul", "Neha"],  
    "Salary": [50000, None, 70000, None],  
    "Age": [25, 28, None, 24]  
}
```

Problems

Count missing values.

Missing Salary ko mean se fill karo.

Missing Age ko median se fill karo.

Rows remove karo jahan Salary missing hai.

Sirf missing values wali rows dikhao.

Level 5: GroupBy

Dataset:

```
data = {  
    "Department": ["IT","HR","IT","Finance","HR","Finance"],  
    "Salary": [50000,45000,70000,65000,48000,72000]  
}
```

Problems

Department-wise average salary.

Department-wise maximum salary.

Department-wise employee count.

Kis department ki average salary highest hai?

Level 6: Data Cleaning

Dataset:

Name

Amit

amit

AMIT

Priya

PRIYA

Problems

Sab names upppercase me convert karo.

Sab names lowercase me convert karo.

Duplicate names remove karo.

Unique names count karo.

Level 7: CSV Practice

28.

CSV file load karo.

```
df = pd.read_csv("employees.csv")
```

```
head()
```

```
tail()
```

```
info()
```

```
describe()
```

run karo.

29.

CSV me:

Missing values count

Duplicate rows count

Highest salary

Lowest salary

nikalo.

30.

CSV ko clean karke save karo.

cleaned_data.csv

Level 8: Interview Style Problems

31.

Second highest salary find karo.

32.

Third highest salary find karo.

33.

Employees jinki salary average salary se zyada hai.

34.

Most common city find karo.

35.

Har city me kitne employees hain.

36.

Top 3 highest paid employees.

37.

Salary aur Age ka correlation nikalo.

38.

Department-wise total salary expenditure.

39.

Employee with maximum age.

40.

Employee with minimum age.